MIL-R-22368D
7 May 1980
SUPERSEDING
MIL-R-22368C
16 July 1975

MILITARY SPECIFICATION

REPAIR KIT AND REPAIR KIT COMPONENTS, FOR

COLLAPSIBLE FABRIC TANKS AND BOATS

This specification is approved for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers a repair kit and the kit components required to temporarily seal holes in collapsible fabric tanks and boats.

2. APPLICABLE DOCUMENTS

2.1 Issues of documents. The following documents of the issue in effect on date of invitation for bids or request for proposal form a part of this specification to the extent specified herein.

SPECIFICATIONS

FEDERAL

MM-L-736	- Lumber; Hardwood.
QQ-A-250/8	- Aluminum Alloy 5052, Plate and Sheet.
PPP-B-601	- Boxes, Wood, Cleated-Plywood.
PPP-B-621	- Boxes, Wood, Nailed and Lock-Corner.
PPP-B-636	- Boxes, Shipping, Fiberboard.

* U.S. GOVERNMENT PRINTING OFFICE: 1980—603-121/2347

FSC 5430

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: US Army Mobility Equipment Research and Development Command, ATTN: DRDME-DS, Fort Belvoir, VA 22060 by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

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MIL-P-116	- Preservation-Packaging, Methods of.
MIL-V-173	- Varnish, Moisture-and-Fungus-Resistant (For Treatment of Communications, Electronic, and Associated Equipment).

STANDARDS

FEDERAL

FED-STD-H28/2	- Screw-Thread Standards for Federal Services, Section 2, Unified Thread Form and Thread Series for Bolts, Screws, Nuts, Tapped Holes and General Applications.

MILITARY

MIL-STD-105	- Sampling Procedures and Tables for Inspection by Attributes.
MIL-STD-129	- Marking for Shipment and Storage.
MIL-STD-1188	- Commercial Packaging of Supplies and Equipment.
MS16535	- Rivet - Tubular, Oval Head.

(Copies of specifications and standards required by contractors in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

2.2 Other publications. The following documents form a part of this specification to the extent specified herein. Unless otherwise indicated, the issue in effect on date of invitation for bids or request for proposal shall apply.

AMERICAN SOCIETY FOR TESTING AND MATERIALS (ASTM)

D 2000 - Rubber Products in Automotive Applications.

(Application for copies should be addressed to the American Society for Testing and Materials, 1916 Race Street, Philadelphia, PA 19103.)

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SOCIETY OF AUTOMOTIVE ENGINEERS (SAE)

SAE Handbook.

(Application for copies should be addressed to the Society of Automotive Engineers, 400 Commonwealth Drive, Warrendale, PA 15096.)

(Technical society and technical association specifications and standards are generally available for reference from libraries. They are also distributed among technical groups and using Federal agencies.)

3. REQUIREMENTS

3.1 Description. The repair kit shall consist of three sizes of repair sealing clamp assemblies (hereinafter called "repair clamp"), and two sizes of wooden plugs in the quantities specified in Table I. Components of the repair kit may be specified separately (see 6.2) when the complete kit is not required.

TABLE I. Repair kit components

Item	Size	Quantity
Clamp	3 inches	4
Clamp	5 inches	2
Clamp	7-1/2 inches	2
Plug	3 inches	2
Plug	5 inches	2
Instructions	-	2

3.2 First article (first produced repair kit). The contractor shall furnish a repair kit, or two repair components of each type and size when kits are not required as specified (see 6.2), for examination and testing within the time frame specified (see 6.2) to prove that his production methods and choice of design detail will produce repair kits that comply with the requirements of this specification. Examination and tests shall be as specified in Section 4 and shall be subject to surveillance and approval by the Government (see 6.3).

3.3 Material. Material shall be as specified herein and as shown on the applicable figures. Materials not specified shall be selected by the contractor and shall be subject to all provisions of this specification (see 6.5).

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3.4 Clamps. Repair clamps shall be as shown in Figure 1. The bottom cup assembly, consisting of the bottom cup, hinge, stud, and rivet, shall withstand a tensile load of not less than 200 pounds when tested as specified in 4.5.2.1. The gasket shall be compounded using only nonmigrating plasticizers and shall have a hardness of not more than 55 Shore A.

3.5 Plugs. The wood plugs shall be as shown in Figure 2. The plugs shall be machined from kiln-dried hardwood lumber, and shall be free from knots, splits, and shakes.

3.6 Instructions. Instructions for use of the repair kit shall be printed on paper and heat-sealed between two sheets of transparent vinyl plastic. The plastic sheets shall be not less than 0.010-inch thickness. The finished instruction sheet shall be no larger in size than 8 inches by 10 inches.

3.7 Identification marking of clamps. The National Stock Number for each clamp (see 6.4) shall be cut permanently on the external surface of the bottom cup in letters not less than 0.125-inch high.

3.8 Workmanship.

3.8.1 Welding. The aluminum surfaces of parts to be welded shall be free from scale, paint, grease, oxide film, and other foreign matter. The welding shall be done by the inert gas, shielded-arc method. Welds may be ground, filed, buffed, or chipped but shall not be hammered, and shall be free from discontinuities such as base metal cracks adjacent to or behind welds, cracks in weld metal, undercutting of base metal, lack of fusion with parent metal or between parts, lack of penetration, lack of prescribed fit, or spatter.

3.8.2 Riveted connections. Rivets shall fill the holes completely. The upset rivet heads shall be full, neatly made, concentric with the rivet holes, and in full contact with the surface of the member, and shall be in accordance with SAE J492.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract, the contractor is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract, the contractor may use his own or any other facilities suit-

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able for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.1.1 Component and material inspection. The contractor is responsible for insuring that components and materials used are manufactured, examined, and tested in accordance with referenced specifications and standards, as applicable.

4.2 Classification of inspections. The inspection requirements specified herein are classified as follows:

(a) First produced repair kit or component inspection (see 4.3).
(b) Quality conformance inspection (see 4.4).
(c) Inspection of packaging (see 4.6).

4.3 First produced repair kit or component inspection.

4.3.1 Examination. The first produced repair kit, or two clamps and two plugs of each size when components are specified separately, shall be examined as specified in 4.5.1. Presence of one or more defect shall be cause for rejection.

4.3.2 Tests. The first produced repair kit, or two clamps of each size when components are specified separately, shall be tested as specified in 4.5.2. Failure of any test shall be cause for rejection.

4.4 Quality conformance inspection.

4.4.1 Sampling. Sampling for examination and tests shall be in accordance with MIL-STD-105, Inspection Level S-1.

4.4.2 Examination. Samples selected in accordance with 4.4.1 shall be examined for the defects specified in 4.5.1. AQL shall be 2.5 percent defective.

4.4.3 Test. Samples selected in accordance with 4.4.1 shall be tested as specified in 4.5.2. AQL shall be 2.5 percent defective.

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4.5 Inspection procedure.

4.5.1 Examination. The repair kits or components shall be examined as specified herein for the following defects:

101. Dimensions not as specified.
102. Materials not as specified.
103. Components missing or not as specified.
104. Identification marking missing, incomplete, or illegible.
105. Instructions missing or not as specified.
106. Workmanship not as specified.

4.5.2 Test.

4.5.2.1 Hinge weld. The bottom cup assembly of the repair clamps shall be tested by applying a static tensile load of not less than 200 pounds to the stud at a right angle to the bottom cup. Separation of the stud, rivet or hinge from the bottom cup, or permanent deformation of the hinge or bottom cup, shall constitute failure of this test.

4.6 Inspection of packaging.

4.6.1 Quality conformance inspection of pack.

4.6.1.1 Unit of product. For the purpose of inspection, a completed pack prepared for shipment shall be considered a unit of product.

4.6.1.2 Sampling. Sampling for examination shall be in accordance with MIL-STD-105, Inspection Level S-1.

4.6.1.3 Examination. Samples selected in accordance with 4.6.1.2 shall be examined for the following defects. AQL shall be 2.5 percent defective.

107. Materials and containers not as specified for Level A or Level B. Each incorrect material or container shall be considered one defect.
108. Clamps not configured as specified.
109. Technical publications not preserved as specified.
110. Marking illegible, incomplete, incorrect, or missing.

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5. PACKAGING

5.1 Preservation. Preservation shall be Level A or Commercial, as specified (see (6.2).

5.1.1 Level A.

5.1.1.1 Clamps. Each clamp shall be preserved in accordance with MIL-P-116, Submethod IC-1. The clamp shall be prepared as follows prior to being sealed in the polyethylene bag:

(a) Unscrew the wingnut from stud.
(b) Remove the top of the clamp and fold the stud down so that it is lying flat inside the bottom cup.
(c) Place the top cup over the bottom and wrap the cord with wingnut around the assembly.

5.1.1.2 Plugs. Each plug shall be preserved in accordance with MIL-P-116, Submethod IC-1.

5.1.1.3 Technical publications. Technical publications shall be preserved in accordance with MIL-P-116, Submethod IC-1.

5.1.1.4 Consolidation. The preserved repair kit consisting of four 3-inch-size clamps, two 5-inch-size clamps, two 7-1/2-inch-size clamps, two 3-inch-size plugs, two 5-inch-size plugs, and two technical publications shall be consolidated in a close-fitting weather-resistant box conforming to PPP-B-636. The box shall be closed in accordance with applicable requirements of the appendix to the box specification.

5.1.2 Commercial. Commercial preservation shall be in accordance with MIL-STD-1188.

5.2 Packing. Packing shall be Level A, Level B, or Commercial, as specified (see 6.2).

5.2.1 Level A. The repair kits, preserved as specified in 5.1, shall be packed in close-fitting boxes conforming to PPP-B-621, Class 2, or PPP-B-601, Overseas Type. Gross weight shall not exceed 200 pounds. The boxes shall be closed and strapped in accordance with the appendix to the applicable box specification.

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5.2.2 Level B. The repair kits, preserved as specified in 5.1, shall be packed as specified in 5.2.1 except that the boxes shall be Overseas Type.

5.2.3 Commercial. Commercial packing shall be in accordance with MIL-STD-1188.

5.3 Marking. In addition to any special marking required by the contract or order, shipping containers shall be marked in accordance with MIL-STD-129 for military levels of protection. Commercial marking shall be in accordance with MIL-STD-1188.

6. NOTES

6.1 Intended use. The repair kit is intended for temporarily repairing collapsible fabric fuel tanks and drums, water tanks, pneumatic boats, life rafts, etc. The components may be specified individually in various quantities when the complete kit is not required.

6.2 Ordering data. Procurement documents should specify the following:

(a) Title, number, and date of this specification.
(b) Number and identity of repair kit components required when the complete kit is not required (see 3.1).
(c) Time frame required for submission of first produced repair kit or components, and number required (see 3.2).
(d) Degree of preservation and degree of packing required (see 5.1 and 5.2).

6.3 First produced repair kit. Any changes or deviations of production repair kits from the approved first produced repair kit during production will be subject to the approval of the contracting officer. Approval of the first produced repair kit will not relieve the contractor of his obligation to furnish repair kits conforming to this specification.

6.4 Cataloging data. For cataloging purposes, the National Stock Number shall be as follows:

3-inch Repair Sealing Clamp Assembly	5340-00-720-8864
5-inch Repair Sealing Clamp Assembly	5340-00-720-8863
7-1/2-inch Repair Sealing Clamp Assembly	5340-00-720-8858
3-inch plug	5510-00-255-9493
5-inch plug	5510-00-255-9492

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6.5 Recycled material. It is encouraged that recycled material be used, when practical, as long as it meets the requirements of this specification (see 3.3).

Custodians:
Army - ME
Navy - YD

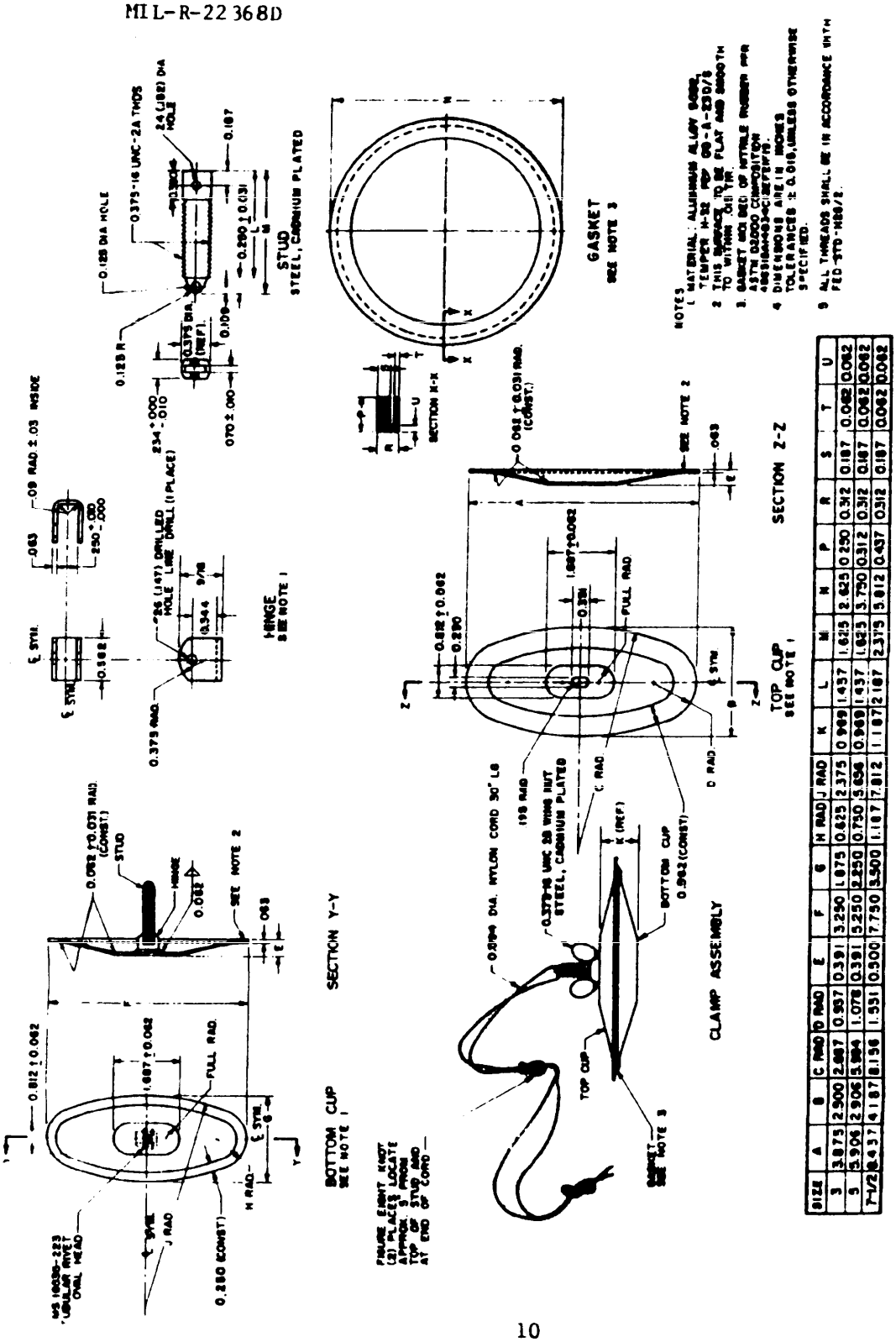
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User activity:
Navy - MC

Preparing activity:
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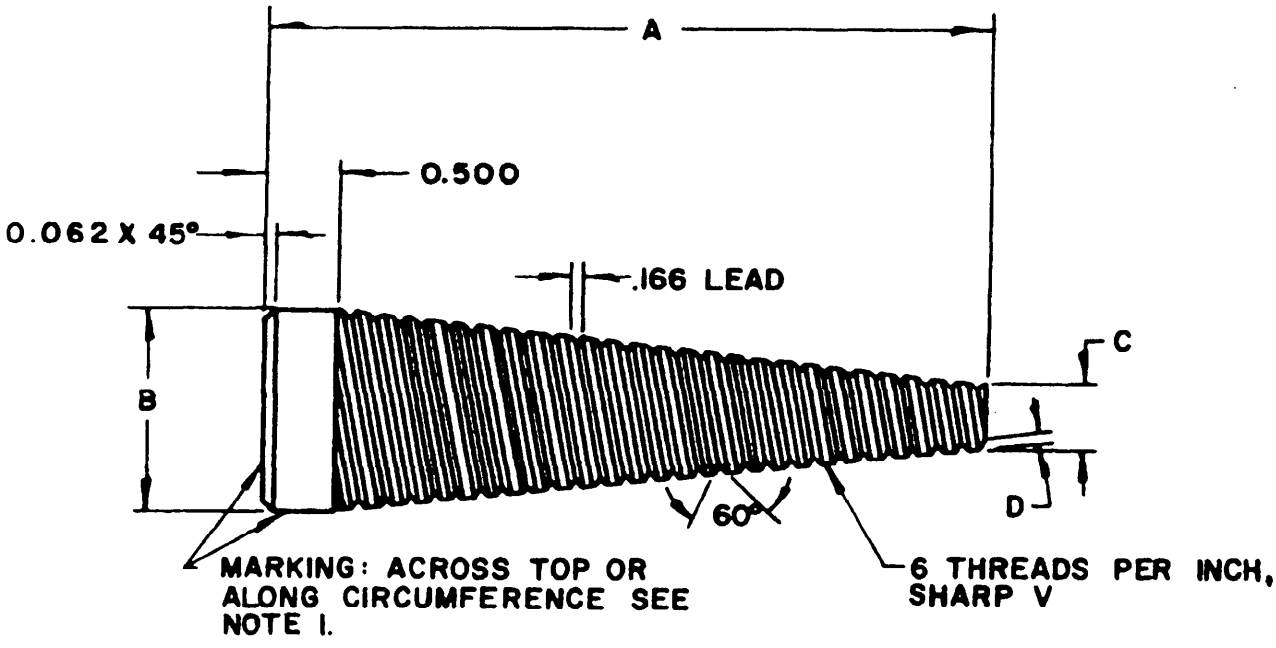


SIZE	A	B	C RAD	D RAD	E	F	G	H RAD	J RAD	K	L	M	N	P	R	S	T	U
3	3.875	2.500	2.887	0.937	0.391	3.250	1.875	0.625	2.375	0.969	1.437	1.625	2.625	0.250	0.312	0.187	0.062	0.062
5	5.906	2.906	3.884	1.078	0.391	5.250	2.250	0.750	3.656	0.969	1.437	1.625	3.750	0.312	0.312	0.187	0.062	0.062
7-1/2	8.437	4.187	5.156	1.531	0.500	7.750	3.500	1.187	7.812	1.187	2.187	2.375	5.812	0.437	0.312	0.187	0.062	0.062

FIGURE 1. Repair sealing clamp assembly.

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SIZE	A	B	C	D
3	3 ± 0.062	0.750 DIA.	0.250 DIA.	0.078
5	5 ± 0.062	1.500 DIA.	0.500 DIA.	0.094

NOTES:

1. IDENTIFICATION MARKING SHALL BE BLACK, INDELIBLE AND WATERPROOF. LETTERS SHALL BE CLEAR AND 0.125 INCH HIGH.
2. AFTER MARKING, PLUG SHALL BE GIVEN ONE COAT OF VARNISH CONFORMING TO MIL-V-173, COMPOSITION I.
3. UNLESS OTHERWISE INDICATED, DIMENSIONS ARE IN INCHES, TOLERANCE ON DECIMALS ± .015, AND ANGLES ± 1° 30'.
4. MATERIAL SHALL CONFORM TO MM-L-736, TYPE I

FIGURE 2. Wood plug, tapered, threaded.

X-2344E

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